

Gregory Macfarlane

Curriculum vitae – Short

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 gregmacfarlane

Post-graduate employment

Brigham Young University

since 08/2026 Associate Professor, *Civil and Construction Engineering*
11/2018-09/2026 Assistant Professor, *Civil and Construction Engineering*

Transport Foundry

05/2017-11/2018 Data scientist

WSP | Parsons Brinckerhoff

10/2016-05/2017 Technical Principal, *Systems Analysis Group*
05/2014-10/2016 Data scientist, *Systems Analysis Group*

Higher education

Georgia Institute of Technology

2014 PhD, *Civil Engineering*
2014 MS, *Economics*

Brigham Young University

2009 BS, *Civil Engineering*, with University Honors

Publications

† BYU undergraduate student ‡ BYU graduate student

Refereed journal articles

- [13] **Macfarlane, G.**, ‡Stucki, E., Redelfs, A. & Spruance, L. (2026). Where's dinner coming from? A utility-based investigation of food access in Utah. *Journal of Transport and Land Use*, 19(1), 249–277. <https://doi.org/10.5198/jtlu.2026.2554>
- [12] ‡Gibbons, N. & **Macfarlane, G. S.** (2025). Evaluating the Impacts of Parameter Uncertainty in a Practical Transportation Demand Model. *Future Transportation*, 5(1), 15. <https://doi.org/10.3390/futuretransp5010015>
- [11] **Macfarlane, G. S.**, ‡Barnes, M. & †Gibbons, N. (2025). A Utility-Based Approach to Modeling Systemic Resilience of Highway Networks with an Application in Utah. *Journal of Transportation Engineering, Part A: Systems*, 151(1), 04024094. <https://doi.org/10.1061/JTEPBS.TEENG-8534>
- [10] **Macfarlane, G. S.**, ‡Riches, G., ‡Youngs, E. K. & Nielsen, J. A. (2024). Classifying Location Points as Daily Activities using Simultaneously Optimized DBSCAN-TE Parameters.. *Findings*. <https://doi.org/10.32866/001c.116197>
- [9] ‡Wang, B., Schultz, G. G., **Macfarlane, G. S.**, Eggett, D. L. & †Davis, M. C. (2023). A Methodology to Detect Traffic Data Anomalies in Automated Traffic Signal Performance Measures. *Future Transportation*, 3(4), 1175–1194. <https://doi.org/10.3390/futuretransp3040064>

- [8] **Macfarlane, G. S.**, ‡Stucki, E., Redelfs, A. H. & Spruance, L. A. (2022). Beyond Proximity: Utility-Based Access from Location-Based Services Data. *International Journal of Environmental Research and Public Health*, 19(19), 12352. <https://doi.org/10.3390/ijerph191912352>
- [7] **Macfarlane, G. S.**, Boyd, N., Taylor, J. E. & Watkins, K. (2021). Modeling the impacts of park access on health outcomes: A utility-based accessibility approach. *Environment and Planning B: Urban Analytics and City Science*, 48(8), 2289–2306. <https://doi.org/10.1177/2399808320974027>
- [6] Glenn, J., †Bluth, M., †Christianson, M., †Pressley, J., Taylor, A., **Macfarlane, G. S.** & Chaney, R. A. (2020). Considering the Potential Health Impacts of Electric Scooters: An Analysis of User Reported Behaviors in Provo, Utah. *International Journal of Environmental Research and Public Health*, 17(17), 6344. <https://doi.org/10.3390/ijerph17176344>
- [5] Brakewood, C., **Macfarlane, G. S.** & Watkins, K. (2015). The impact of real-time information on bus ridership in New York City. *Transportation Research Part C: Emerging Technologies*, 53, 59–75. <https://doi.org/10.1016/j.trc.2015.01.021>
- [4] **Macfarlane, G. S.**, Garrow, L. A. & Moreno-Cruz, J. (2015). Do Atlanta residents value MARTA? Selecting an autoregressive model to recover willingness to pay. *Transportation Research Part A: Policy and Practice*, 78, 214–230. <https://doi.org/10.1016/j.tra.2015.05.010>
- [3] **Macfarlane, G. S.**, Garrow, L. A. & Mokhtarian, P. L. (2015). The influences of past and present residential locations on vehicle ownership decisions. *Transportation Research Part A: Policy and Practice*, 74, 186–200. <https://doi.org/10.1016/j.tra.2015.01.005>
- [2] Binder, S., **Macfarlane, G. S.**, Garrow, L. A. & Bierlaire, M. (2014). Associations among household characteristics, vehicle characteristics and emissions failures: An application of targeted marketing data. *Transportation Research Part A: Policy and Practice*, 59, 122–133. <https://doi.org/10.1016/j.tra.2013.11.005>
- [1] Wall, T. A., **Macfarlane, G. S.** & Watkins, K. E. (2014). Exploring the Use of Egocentric Online Social Network Data to Characterize Individual Air Travel Behavior. *Transportation Research Record*, 2400(1), 78–86. <https://doi.org/10.3141/2400-09>

Peer-reviewed conference papers

- [3] **Macfarlane, G. S.**, ‡Day, C., Erhardt, G. D. & Watkins, K. E. (2026). Identifying ridehail riders through pairing ActivitySim and BEAM. In *Procedia Computer Science* (pp. 286–293). <https://doi.org/10.1016/j.procs.2026.04.038>
- [2] ‡Jarvis, D. L., **Macfarlane, G. S.**, †Woolley, B. & Schultz, G. G. (2024). Simulating Incident Management Team Response and Performance. In *Procedia Computer Science* (pp. 91–96). <https://doi.org/10.1016/j.procs.2024.06.002>
- [1] **Macfarlane, G. S.** & ‡Lant, N. (2023). How Far Are We From Transportation Equity? Measuring the Effect of Wheelchair Use on Daily Activity Patterns. In C. Antoniou, F. Busch, A. Rau & M. Hariharan (Eds.), *Proceedings of the 12th International Scientific Conference on Mobility and Transport* (pp. 141–155). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-8361-0_10

Conferences & invited talks

Invited talks

- 2025 California HSR model transferability, CalTrans Transit Research-to-Practice Symposium
- 2023 Modeling with Big Data, course presentation, Technische Universität München

2020 Using Mobile Data to Improve Transportation Sustainability, Utah Central ASCE Chapter luncheon

Courses

CCE 102: Sustainable Infrastructure

Sustainability in built environments and infrastructure with a focus on global challenges and solutions. Converted from CCE 201 in Fall 2024.

CE 361: Introduction to Transportation Engineering

Transportation systems characteristics, traffic engineering and operations, transportation planning, geometric design, pavement design, transportation safety, freight, public transport, sustainable transportation.

CE 565: Urban Transportation Planning

Characteristics of urban transportation planning and decision making, intermodal transportation, land-use transportation interrelationships, transportation demand modeling, site impact analysis, sustainable transportation, and livable cities.

CE 662: Transport Simulation and Analysis

An advanced graduate course in traffic flow theory and simulation. Topics include shock wave analysis, discrete event simulation of queues and daily activity pattern choices, car following models, and traffic simulation. Laboratory assignments use MATSim and PTV Vissim simulation software.

Awards & honors

- 2023 **TUM Global Visiting Professor.** Selected to enrich the research culture at Technische Universität München through innovative approaches and exploration of new, cutting-edge research fields.
- 2022 **Most Influential Faculty.** Given to the Civil Engineering faculty member whom graduating seniors named as the most influential on their undergraduate education.
- 2022 **ASCE ExCEED Teaching Fellow.** Participated in a week-long intensive teacher development program.

Citizenship & service

External citizenship

- 2023–present **NCHRP 08-184:** Peer review panel member. Framework for Assessing Induced Demand Effects of Various Roadway Investments
- 2019–2025 **Transportation Research Board, AEP50: Travel Demand Forecasting:** Committee member; chair, travel forecasting resources subcommittee; editor of tfresource.org

Professional memberships

- 2022–present American Society of Civil Engineers
- 2020–2022 Zephyr Foundation
- 2009 Tau Beta Pi. Utah β '09

Internal citizenship

2023–present **Department undergraduate committee:** Co-chair. Member since 2021. Lead curriculum revisions for the civil engineering program and ABET accreditation and continuous-improvement efforts.